26-ES-CS201 — Programming for Problem Solving and Introduction to AI

Course Code	26-ES-CS201	**Category**	Engineering Science Course
Course Title	Programming for Problem Solving and Introduction to AI	**L – T – P**	3 – 0 – 0
Credits	3	**Contact Hours**	36
Semester	Second	**Assessment**	Theory: CA 30 + ESE 70

Prerequisites

A three-credit course of thirty-six hours, following the syllabus adopted by the University and carrying, in Module VIII, an introduction to Artificial Intelligence. It assumes no prior programming. Assumed:

- Computer Science or Informatics Practices of the Higher Secondary course where taken (CBSE XI–XII; WBCHSE Computer Application); otherwise nothing.
- The problem-solving mathematics of Mathematics-I.
- Where much of the class has programmed before, Modules I–II are compressed and the time given to Modules VI–VIII.

Course Objectives

- To develop computational thinking — decomposition, abstraction and algorithm design.
- To establish programming in C from first principles to pointers, structures and files.
- To introduce the analysis of an algorithm's efficiency.
- To introduce Artificial Intelligence — its concepts, types and applications — as the foundation for later study.

Course Outcomes

On completion of the course the student will be able to:

CO	Statement	Bloom
CO1	Explain the organisation of a digital computer and the translation of a high-level program.	L2 – Understand
CO2	Apply computational thinking — decomposition, abstraction and algorithm design — to a problem and express the solution as an algorithm and flowchart.	L3 – Apply
CO3	Write, test and debug C programs using control structures, functions, arrays, strings and pointers.	L3 – Apply
CO4	Use structures, files and the elementary search and sort algorithms, and reason about an algorithm's efficiency.	L4 – Analyse
CO5	Explain the concepts, types and applications of Artificial Intelligence and the distinction between human and machine intelligence.	L2 – Understand

CO–PO Articulation Matrix

Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	2	–	–	–	–	–	2
CO2	3	3	2	2	2	–	–	–	–	–	2
CO3	3	3	3	2	3	–	–	1	–	–	2
CO4	3	3	2	3	3	–	–	–	–	–	2
CO5	2	2	2	1	2	1	1	–	1	–	3

Course Plan — Module-wise Syllabus

Module	Title	Contents	Hours
I	Introduction to Computer	Information and data; the components of a digital computer — the CPU, primary and secondary storage, and input–output devices; system and application software; machine language, assembly language and high-level languages.	4

Module	Title	Contents	Hours
II	Generation of Computers	Super, mainframe, mini and personal computers; high-performance computing systems; graphics-processing-unit (GPU) computers.	2
III	Problem Solving and Computational Thinking	The stored-program idea; compiler, interpreter and the translation process; problem definition, decomposition, abstraction and pattern recognition; the algorithm and its properties; flowchart symbols and construction; pseudocode; stepwise refinement; testing an algorithm by hand tracing; the classification of errors — syntax, run-time and logical.	4
IV	Fundamentals of C	Structure of a C program; character set, identifiers and keywords; data types and qualifiers; constants and variables; operators, precedence and associativity; type conversion; formatted and unformatted input and output; conditional statements — if, if–else, nested if and switch; loops — while, do–while and for; break, continue and goto; nested loops.	5
V	Functions and Program Structure	The function as a unit of abstraction; declaration, definition and call; parameter passing by value; return values; scope and lifetime; storage classes; recursion and its trace — factorial, Fibonacci, greatest common divisor and the Towers of Hanoi; recursion compared with iteration; the preprocessor, macros and header files; modular design and separate compilation.	5
VI	Arrays, Strings and Pointers	One- and two-dimensional arrays — declaration, initialisation, traversal and passing to a function; matrix operations. Strings and the standard string library. Pointers — the address and dereference operators, pointer arithmetic, pointers and arrays, pointer to a function, array of pointers; call by reference; dynamic memory allocation with malloc, calloc, realloc and free; common pointer errors.	6
VII	Structures, Files and Algorithm Efficiency	Structures and unions; array of structures; nested structures; pointer to a structure; typedef and the enumerated type. File handling — text and binary streams, opening modes, reading and writing, random access and error handling; reading engineering data from a comma-separated file. Linear and binary search; bubble, selection and insertion sort; the order of growth and elementary comparison of algorithms.	4
VIII	Introduction to Artificial Intelligence	What Artificial Intelligence is and how it works; its advantages and disadvantages; a brief history; types of AI — weak and strong AI; machine learning and deep learning in outline. Defining intelligence and its components; the difference between human and machine intelligence; the agent and its environment; different applications of AI.	6

Cross-cutting notes. The concurrent laboratory (ES-CS291, Programming for Problem Solving Lab & AI Tools) carries the practice, including the AI-tools exercises that accompany Module VIII. The course is the entry point of the computational thread that runs through every programme, and Module VIII opens the artificial-intelligence content that the later years develop.

Text and Reference Books

- Venkatesh, Nagaraju Y, Practical C: Programming for Problem Solving, Khanna Publishing House.
- E. Balagurusamy, Programming in ANSI C, McGraw-Hill.
- Kernighan and Ritchie, The C Programming Language, Pearson.
- Byron Gottfried, Programming with C (Schaum's Outline), McGraw-Hill.
- Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach (introductory chapters), Pearson.
- Reema Thareja, Programming in C, Oxford University Press.

Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Problem Solving through Programming in C (IIT Kharagpur), 12 weeks.
- NPTEL — Introduction to Programming in C, 8 / 12 weeks.
- NPTEL — An Introduction to Artificial Intelligence / AI: Search Methods for Problem Solving, 12 weeks.
- SWAYAM — Programming in C; Introduction to AI, as offered.

Question-paper note. At least half the marks of Groups B and C require the writing or reading of C code. At least one Group C question is on Module VIII (Artificial Intelligence). Dry-run and output-prediction questions are encouraged in Group A.